

E-Commerce Platform - Low-Level Design (LLD)

1. Tech Stack (Suggested)

- **Frontend:** React / Next.js
- **Backend:** Node.js (Express / NestJS)
- **Database:** MongoDB / PostgreSQL (based on use-case)
- **Authentication:** JWT + Refresh Tokens
- **Storage:** AWS S3 (for product images, etc.)
- **Search:** Elasticsearch (for fast product search)
- **Caching:** Redis (for cart, frequently viewed items)
- **Queueing:** RabbitMQ / BullMQ (for async tasks like order confirmation emails)

2. User Management

2.1 Tables / Collections

- User: { id, name, email, phone, passwordHash, role, address[], createdAt }
- Address: { id, userId, line1, line2, city, state, zip, country, isDefault }
- Wishlist: { id, userId, productId, createdAt }

2.2 Features

- Signup / Login
- Forgot Password
- Profile Update
- Address Management
- Role: Customer, Admin, Seller
- Wishlist / Like System

3. Product & Catalog Management

3.1 Tables / Collections

- Category: { id, name, parentCategoryId }
- Product: { id, name, description, brand, categoryId, price, images[], specs[], createdAt }
- ProductVariant: { id, productId, size, color, stock, sku, price }
- Inventory: { variantId, quantityAvailable }

3.2 Features

- Multi-level categories
- Product creation & edit (Admin/Seller)
- Variant-based stock management
- Search & Filter
- Brand filtering

- Discount tags, labels (e.g. "Bestseller")

4. Cart Management

4.1 Tables / Collections

- Cart: { userId, items[] }
- CartItem: { variantId, quantity }

4.2 Features

- Add to cart
- Update quantity
- Remove from cart
- Guest cart (localStorage / cookie)

5. Order Management

5.1 Tables / Collections

- Order: { id, userId, totalAmount, status, createdAt, paymentStatus }
- OrderItem: { orderId, variantId, quantity, price }
- Shipment: { orderId, courier, trackingId, status, estimatedDelivery }

5.2 Features

- Order creation
- Track order
- Status: Pending, Confirmed, Shipped, Delivered, Cancelled
- Auto-update via cron/queue

6. Payment Management

6.1 Tables / Collections

- Payment: { orderId, amount, method, status, paymentGatewayId, createdAt }

6.2 Features

- Integration with Razorpay/Stripe
- Success & Failure handling
- Refunds (basic support)
- COD (Cash on Delivery)

7. Review & Rating

7.1 Tables / Collections

- Review: { id, userId, productId, rating, comment, createdAt }

7.2 Features

- Product-wise ratings
- Display average rating
- Report inappropriate reviews

8. Search & Discovery

Features

- Full-text search (Elasticsearch)
- Filters: Brand, Price, Category
- Sort: Popularity, Price, Rating
- Auto-suggest search bar

9. Admin / Seller Panel

9.1 Features

- Product CRUD
- Order dashboard
- Revenue dashboard
- Inventory alerts
- Banner and Promotion Management
- Bulk upload via CSV

10. Security

Features

- JWT-based auth
- Rate limiting
- XSS, CSRF protection
- Input validation

11. Notifications

Channels

- Email (via nodemailer/sendgrid)
- SMS (Twilio)
- Push Notifications (Firebase or Web)

Use Cases

- Order confirmation

- Shipment updates
- Promotional messages

12. Analytics & Reporting

Metrics

- Sales per category
- Top-selling products
- Abandoned cart rate
- Daily orders/revenue

13. Testing Strategy

- Unit Tests (Jest)
- Integration Tests
- Load Testing (k6 / Artillery)
- E2E (Cypress)

14. DevOps & Deployment

- CI/CD: GitHub Actions / GitLab CI
- Docker-based containerization
- Hosting: Vercel (frontend) + AWS/DigitalOcean (backend)
- Monitoring: LogRocket / Sentry / Datadog

Frontend Pages (React / Next.js)

1. Public Pages

Page	Route	Description
Home	/	Hero banners, category highlights, trending products
Category Listing	/category/:slug	Shows products under a category (e.g., Electronics)
Product Details	/product/:slug	Individual product info with variants, reviews, etc.
Search Results	/search?q=...	Search page with filters, sort
Cart	/cart	View/update cart items
Checkout	/checkout	Address, payment, review order
Order Confirmation	/order-confirmation/:id	Thank-you page after order
Login	/login	User login form
Signup	/signup	User registration form
Forgot Password	/forgot-password	Email form to reset password
Reset Password	/reset-password/:token	New password form
Profile	/profile	Profile + address mgmt
Orders	/orders	List of past orders
Order Details	/orders/:id	Order invoice & tracking
Contact Us	/contact	General support
About Us	/about	Company info
Terms & Privacy	/terms, /privacy	Legal pages

2. Admin/Seller Pages

Page	Route	Description
Admin Dashboard	/admin	Stats, sales summary
Products Management	/admin/products	List/create/update products
Add Product	/admin/products/new	Product form
Edit Product	/admin/products/:id/edit	Edit existing product
Orders Dashboard	/admin/orders	All placed orders
View Order	/admin/orders/:id	Order details, shipping status
Categories Management	/admin/categories	Add/edit/delete categories
Users Management	/admin/users	View users
Banners / Promotions	/admin/banners	Manage homepage banners
Reviews Moderation	/admin/reviews	Flag/delete reviews
Support Tickets	/admin/support	View/respond to user tickets

Backend API Routes

All routes are prefixed with /api

Auth

Method	Route	Description
POST	/api/auth/signup	Register user
POST	/api/auth/login	Login user
POST	/api/auth/logout	Logout user
POST	/api/auth/forgot	Send reset email
POST	/api/auth/reset	Reset password
GET	/api/auth/me	Get current user info

Products & Categories

Method	Route	Description
GET	/api/products	Get all products
GET	/api/products/:id	Get one product
POST	/api/products	Add new product (admin)
PUT	/api/products/:id	Update product
DELETE	/api/products/:id	Delete product
GET	/api/categories	Get all categories
POST	/api/categories	Add category
PUT	/api/categories/:id	Edit category
DELETE	/api/categories/:id	Delete category

Cart

Method	Route	Description
GET	/api/cart	Get user cart
POST	/api/cart	Add item to cart
PUT	/api/cart/:variantId	Update quantity
DELETE	/api/cart/:variantId	Remove item

Checkout & Orders

Method	Route	Description
POST	/api/checkout	Initiate checkout (address, payment, items)
GET	/api/orders	Get all orders (user)
GET	/api/orders/:id	Get single order
POST	/api/orders	Place new order
PATCH	/api/orders/:id	Update status (admin)

Payment

Method	Route	Description
POST	/api/payment/initiate	Start payment (Razorpay/Stripe)
POST	/api/payment/verify	Verify payment callback

Reviews

Method	Route	Description
GET	/api/products/:id/reviews	Get reviews for a product
POST	/api/products/:id/reviews	Add a review
DELETE	/api/reviews/:id	Delete review (user or admin)

Admin Specific

Method	Route	Description
GET	/api/admin/dashboard	Stats overview
GET	/api/admin/orders	All orders
GET	/api/admin/products	All products
GET	/api/admin/users	All users
POST	/api/admin/banners	Manage homepage banners
GET	/api/admin/support	View support tickets